

João Rietra

AI Engineer / Data Scientist · model evaluation, data architecture and applied AI

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SUMMARY

AI Engineer and Data Scientist with a medical background, creator of an out-of-domain model evaluation standard that I apply in every team I join. Tested in my Master's dissertation and replicated in every model I ship. I measure calibration and out-of-distribution generalisation as acceptance criteria. I design the data architecture that underpins production, from the relational store to the vector store. I run production systems: neural networks detecting image manipulation, an LLM serving real users, and ML models auditing the power grid to find energy losses at national scale.

KEYWORDS

Data Science · Python · PyTorch · scikit-learn · Hugging Face Transformers · Pandas · NumPy · DuckDB · SQL · ETL/ELT · MLflow · model calibration (ECE) · out-of-distribution evaluation (OOD) · time series and foundation models · inferential statistics (CI, Wilcoxon) · dashboards (Streamlit) · Brazilian public data (IBGE, ONS, INEP) · computer vision · OCR · Docker · AWS · LGPD and sensitive data · technical mentoring and teaching · subproject technical leadership · cross-functional teamwork · agile methodologies (Scrum/Kanban)

EDUCATION

- **Professional Master's in Software Engineering** — CESAR School, Recife · Oct 2025 – defence expected Nov 2026
 - **Technologist Degree in Data Science** (3-year undergraduate) — CESAR School, Recife · Jan 2025 – expected completion Jul 2027
 - **Doctor of Medicine (MD)** — FPS / IMIP, Recife · completed 2022
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EXPERIENCE

AI Engineer — LetzGrow (School Innovation, CESAR School) · 2026 – present

AI-assisted career platform: candidates move through a guided journey with an AI mentor and receive a personalised development plan. Selected for 2026.2, as a two-person team.

- **Candidate↔opportunity matching** refined through a double opt-in loop, shortening time to match.
- **Fairness evaluation** of the recommendation pipeline, reported in a versioned document.
- **Sensitive data:** conditional access and audit trail aligned with Brazil's LGPD (GDPR-equivalent), on PostgreSQL 15 with pgvector semantic search.

Technical Lead (data and computer vision) — MAPEN (School Innovation, CESAR School) · 2026 – present

System that locates energy losses by comparing distributed against consumed energy, with red flags by region. Selected for the 2026.2 edition. My scope: data and computer vision.

- **Nationwide data layer:** hourly grid-load series from Brazil's national grid operator (2000→2026), billed consumption by state and IBGE boundaries in one queryable DuckDB base. The gap between measured load and billed consumption is the loss signal itself.
- **Geospatial-temporal anomaly detection:** modelling the expected vs. observed gap by area, generating red flags prioritised by criticality.
- **Automatic meter-display reading:** fine-tuned TrOCR-small on the UFPR-AMR dataset, benchmarked against PP-OCRv6-tiny, with MLflow tracking.

ML Engineer — Veritas (School Innovation, CESAR School) · 2025 – 2026

Deepfake detection: computer vision with Grad-CAM and OOD generalisation testing; forensic score combined with a generative-AI explanation.

Teaching Assistant — CESAR School · 2025 – 2026

- Programming (C), Statistics (R), Calculus I, Competitive Programming and Data Analysis and Visualization.

Medical Data Analyst — J & L Hospital Consulting · 2023 – 2024

Analytics for hospital care and management, combining clinical and managerial data.

- **Clinical outcomes:** measured readmission, hospital-acquired infection and post-operative complications, pointing out failures and supporting protocol adjustments.
- **Quality and accreditation:** tracked the KPIs required by the ONA and JCI accreditation bodies.
- **Operations and costs:** analysed bed occupancy, operating-room productivity, inpatient costs and tariff deviations, with Streamlit dashboards for hospital leadership.

ACADEMIC OUTPUT

1. *Training Reuse for Binary Triage in Multi-Domain Medical Imaging* (Master's dissertation, CESAR School, 2026; title in Portuguese: *Reuso de Treino para Triagem Binária em Imagem Médica Multidomínio*): controlled comparison of 4 backbone origins across 5 domains (MedMNIST), evaluating AUC, calibration (ECE) and occlusion-based explanation faithfulness. Forensic intermediate fine-tuning won (AUC 0.842 vs. 0.796 for ImageNet) with far better calibration (ECE 0.045 vs. 0.219). MLflow and statistical aggregation (CI, Wilcoxon) over multiple seeds.
2. *Asymmetric generalisation in image-manipulation detectors: from the training domain to collapse on manual splicing* (SBTI 2026, submitted; Portuguese title: *Generalização assimétrica em detectores de manipulação de imagens: do domínio de treino ao colapso em splicing manual*).
3. *Fine-tuning decision forests under drift: a Bayesian blend with a unified vocabulary* (SBTI 2026, submitted; Portuguese title: *Fine-tuning de decision forests na presença de drift: um blend bayesiano com vocabulário unificado*): DeepJungle library, validated on 4 public benchmarks (Folktables/ACS, Electricity, Covertypes, rotated MNIST). github.com/jhlr/deepjungle
4. *EfficientNeXt: an autoregressive mini-LLM derived from the EfficientNet architecture with unified generation and embedding (GRIT) and decoding steering without an auxiliary model* (Portuguese original: *EfficientNeXt: um mini-LLM autoregressivo derivado da arquitetura EfficientNet com geração e embedding unificados (GRIT) e steering de decoding sem modelo auxiliar*).
5. *Automated inspection of electricity-meter reading photos: an OCR pipeline and characterisation of the coherence-validation problem* (SBTI 2026, submitted; Portuguese title: *Fiscalização automatizada de fotos de leitura de medidores de energia elétrica: um pipeline de OCR e caracterização do problema de validação de coerência*).

PROJECTS

btc-forecast — foundation models and fine-tuning · github.com/jhlr/BTC-Forecast

Systematic comparison of 5 zero-shot time-series foundation models (Chronos, TimesFM, Moirai, Lag-Llama, TTM) with fine-tuning, ensembling and directional backtesting, tracked in MLflow.

desemprego-educacao-brasil · github.com/jhlr/Desemprego-Brasil

Education (IDEB/INEP) × unemployment (PNAD/IBGE via the SIDRA API): EDA, Streamlit dashboard and a PDF report.

SKILLS

Machine learning: PyTorch · scikit-learn · Hugging Face Transformers · OpenCV · transfer learning and fine-tuning (LoRA) · OCR (TrOCR, PaddleOCR) · Grad-CAM and explainability · calibration (ECE) · OOD evaluation · anomaly detection · time series and foundation models

Data: Pandas · NumPy · DuckDB · PostgreSQL and SQL · ETL/ELT · geospatial data (GeoJSON) · Brazilian public data (IBGE, ONS, INEP) · inferential statistics (CI, Wilcoxon) · dashboards (Streamlit)

MLOps, generative AI and domain: MLflow · Docker · AWS · Git · LLMs (OpenAI, Anthropic, Groq, Gemini, Ollama) · RAG and vector search · prompt engineering · LGPD and sensitive data · clinical domain modelling

Languages: Portuguese (native) · English (native) · Spanish (basic) · French (basic)